

THE SUPER-YANGIAN SHADOW TOWER: RICCATI EXPANSION AND THE $\kappa_{\text{ch}} + \kappa_{\text{ch}}^! = \max(m, n)$ COMPLEMENTARITY

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ABSTRACT. SCAFFOLD STUB. For super-Yangian inputs $Y(\mathfrak{sl}(m|n))$, the complementarity identity reads $\kappa_{\text{ch}}(Y(\mathfrak{sl}(m|n))) + \kappa_{\text{ch}}(Y(\mathfrak{sl}(n|m)))^! = \max(m, n)$ (scope: Sugawara-shifted dual level; corrects the naive sum-o analogy; retraction B86). The super-Riccati expansion yields closed-form shadow coefficients S_r with characteristic arithmetic primes distinct from the bosonic Virasoro tower: the two pairings (super-trace vs Berezinian) coexist without programme-level canonicalisation pending Verdier-pairing inscription (frontier item).

PRIMARY THEOREMS AND STATUS

- $\kappa_{\text{ch}}(Y(\mathfrak{sl}(m|n))) + \kappa_{\text{ch}}(Y(\mathfrak{sl}(n|m)))^! = \max(m, n)$: (symbolically verified at small rank; replaces naive = o analogy).
- Super-Riccati closed forms for S_r through $r = 8$: .
- Canonical Verdier pairing (super-trace vs Berezinian canonicalisation): (open frontier per 2026-04-17 Beilinson audit).

CHAPTER INPUTS (POINTERS)

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\input{../chapters/examples/super_riccati_shadow_tower_platonic.tex}
\input{../chapters/theory/quantum_groups_foundations.tex}
\input{../chapters/theory/quantum_chiral_algebras.tex}
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CROSS-VOLUME DEPENDENCIES

- Vol I riccati.tex standalone (bosonic Riccati algebraicity and depth classification).
- Vol I shadow-tower higher-coefficients theorem (thm: shadow-series-closed-form-Virasoro).
- Vol II tempering bound β_N (Vol II CLAUDE.md B81).

RETRACTIONS INSCRIBED

B86: the naive super-Yangian sum-vanishing complementarity analogy (the incorrect Virasoro-styled identity with right-hand side zero) is retracted in favour of the Sugawara-shifted dual identity with right-hand side $\max(m, n)$ (scope: super-Yangian of $\mathfrak{sl}(m|n)$; the Virasoro-style 13-constant is a distinct family). B87: the “tempered stratum obstruction $\kappa_{\text{ch}}^{(\infty)}_{\text{orig}} = 1/e$ ” claim retracted (Stirling cancellation error); correct $\limsup = 0$ at generic c .